

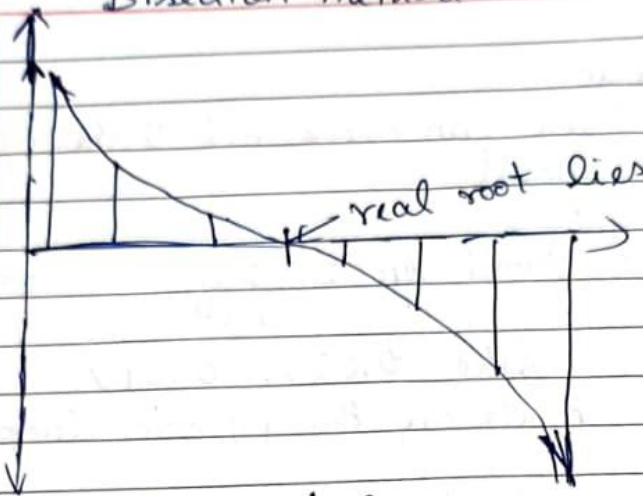
Bisection method

Date	
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Made by F

- Aniket Singh

Roll no: 8824103100



→ what we do?

substitution/hit and trial

$f(x), f(1), f(2) \dots$

$x=1, 2, 3 \dots$

also continuous

~~if~~ what we do?

if $f(x) \times f(y) < 0$ (negative)

we will get average

$f\left(\frac{x+y}{2}\right)$

and then search

$$x^3 + x - 7$$

$$f(0) = 0 + 0 - 7 = -7 \quad \text{multiplication is } > 0$$

$$f(1) = 1 + 1 - 7 = -5 \quad \text{multiplication } < 0$$

$$f(2) = 8 + 1 - 7 = 2$$

~~over~~ root lies btw

$f(1)$ and $f(2)$

$$f\left(\frac{1+2}{2}\right) = f(1.5) = 3.375 - 7 = -3.625$$

root lies between

$f(1.5)$ and $f(2)$

computer logic

error → we can check till 2 decimal point

error % should lie in 2 decimal ~~set~~ something

like 0.62%, 0.01%.

used for closing the infinite loop

~~error~~ = current iteration

$$\text{error \%} = \frac{P.I - C.I}{C.I} \times 100$$

Computer logic / approach

we will iterate using while loop till we find roots.

while (f(x) != 0 ?), ~~no~~ something like this

wrong ~~f(x) > 0~~

[while ((b-a) >= ~~tolerance~~ error)]

error is range where inaccuracy is allowed

if (f(a) * f(b)) > 0) # should be

negative or return

C = (a+b)/2, average

Assuming a is +ve

now, the thing is

if (f(c) == 0) {

return C;

}

else if (f(c) * f(a) < 0)

b = c;

}

else {

a = c

}

we will start with, 0,

a=0

b=1,

now, ~~if~~ using while loop

~~if~~ ~~(foundroot)~~

val1 = ~~a~~ f(a);

val2 = f(b);

~~error~~ = $\left(\frac{\text{val1} - \text{val2}}{\text{val2}} \right) \times 100$

error = (val1 - val2) / val2

~~b = a~~; ~~bisection-root~~
~~b += 1~~; (if f(a) * f(b) < 0)

can switch with val1, val2

root = ~~b~~ bisection-root(a, b, error);
~~break~~;

cout << ~~root~~ root;

Notes
for
Authors

The notes looks
~~awful~~
awful, but worth
a try.

[Update]

Code was flawless, ran in 0.885
seconds.